

Module 3: Strategic Equilibrium: Foundations & Applications

ECO364-THINKING STRATEGICALLY

Part III: Nash Equilibrium

APPLICATIONS

Traffic Network Design

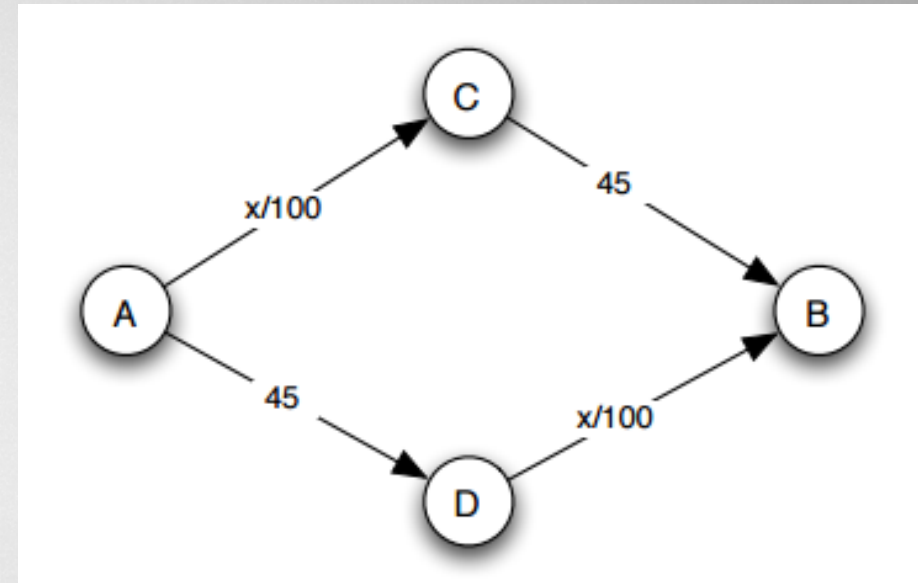
TRAFFIC FLOWS AND BRAESS' PARADOX

Traffic at Equilibrium

- ▶ In this part, we will apply our knowledge of Nash Equilibrium to a model of a transportation network and we will see how it responds to traffic congestion.
- ▶ We will represent a transportation network by a directed graph:
 - ▶ Edges are the roads or highways.
 - ▶ Nodes are exists where you can get on or off a highway.
 - ▶ Each road or highway has attached a “congestion measure” which we will interpret as how much time it takes to traverse that particular road as a function of the amount of traffic. You can think of the congestion measure as the travel time as a function of the amount of traffic the road contains.

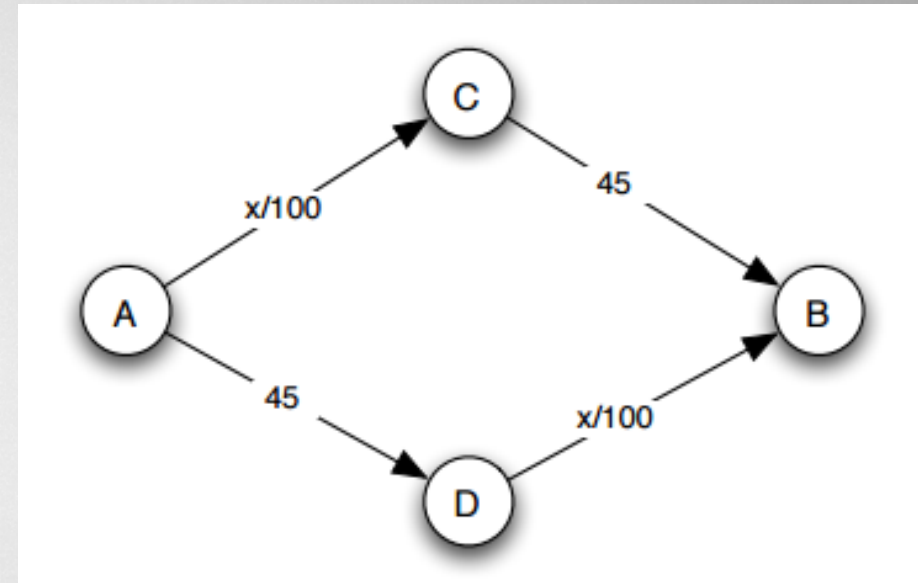
Traffic network: Example

- ▶ Consider the traffic network in the figure.
- ▶ The label in each edge gives the travel time in minutes when there are x cars using the edge.
- ▶ In this example edges AD and CB are insensitive to congestion: each takes 45 minutes to traverse regardless of the number of cars.
- ▶ On the other hand, edges AC and DB take each $x/100$ minutes to traverse when there are x cars using the edge.
- ▶ Suppose that the total traffic flow is 4000 cars: this amount of cars depart from A and want to reach B.
- ▶ There are two possible routes that each car can take: route ACB or route ADB.
- ▶ For example, if all cars take route ACB then total travel time is $4000/100+45=85\text{min}$.
- ▶ The same if all cars take route ADB.
- ▶ If the cars divide evenly between routes ACB and ADB, so that each route carries 2000 cars, then total travel time for people on both routes is $2000/100+45=65$.



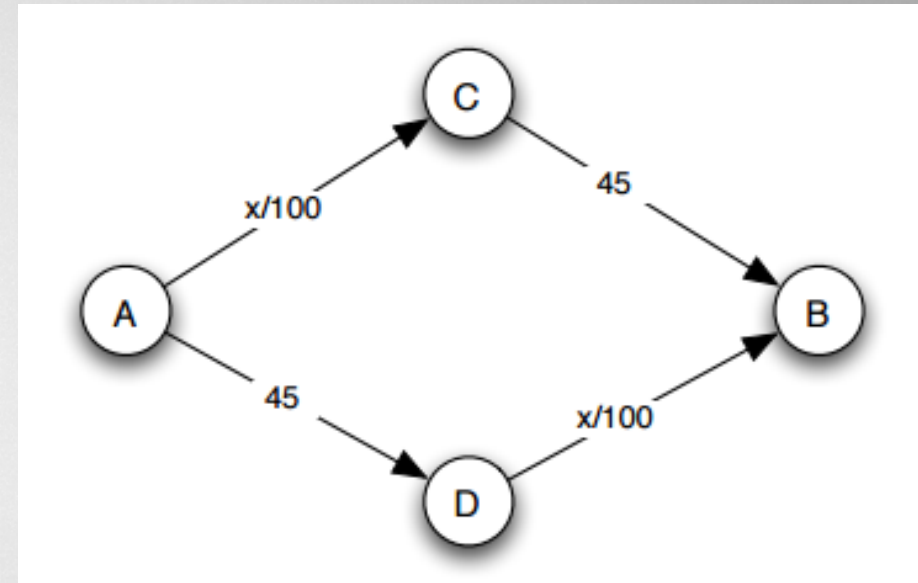
Traffic equilibrium

- ▶ The traffic model I described is a game in which players correspond to drivers (4000 players), strategies are the routes (ACB or ADB) and the payoff for a player is the negative of his or her travel time. We of course are assuming that large travel times are bad.
- ▶ Although we have focused on games with a few players whereas in this game the number of players is very large, the analysis remains simple.
- ▶ An equilibrium is an specification of a route, one for each driver.
- ▶ To characterize the equilibria (because there are many), we will use the fact that in a Nash Equilibrium, there should be no optimal deviation.



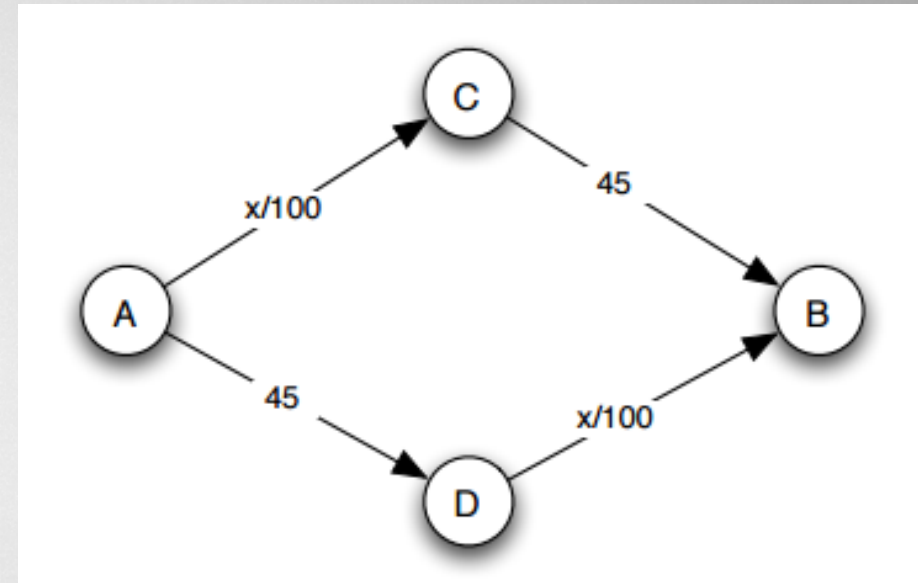
Traffic equilibrium

- ▶ When does a driver have an optimal deviation? Think about it. Let's say that you are driving through route ACB and given the amount of traffic in this route, it would take you 75 minutes to traverse it, whereas in the alternative route ADB it would take you 55 minutes.
- ▶ You would of course change to route ADB, because it is an optimal deviation.
- ▶ Thus, for a traffic pattern (specification of a route for each player) to be an equilibrium, travel times for the two routes, ACB and ADB, must be equal.
- ▶ In this way, there is no optimal deviation and this traffic pattern would be an equilibrium!



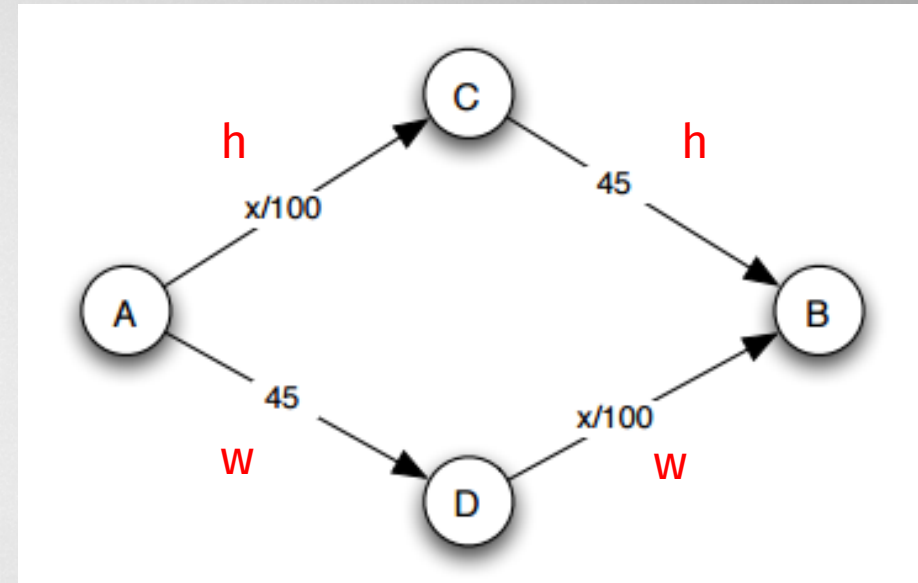
Traffic equilibrium

- ▶ When does a driver have an optimal deviation? Think about it. Let's say that you are driving through route ACB and given the amount of traffic in this route, it would take you 75 minutes to traverse it, whereas in the alternative route ADB it would take you 55 minutes.
- ▶ You would of course change to route ADB, because it is an optimal deviation.
- ▶ Thus, for a traffic pattern (specification of a route for each player) to be an equilibrium, travel times for the two routes, ACB and ADB, must be equal.
- ▶ In this way, there is no optimal deviation and this traffic pattern would be an equilibrium!



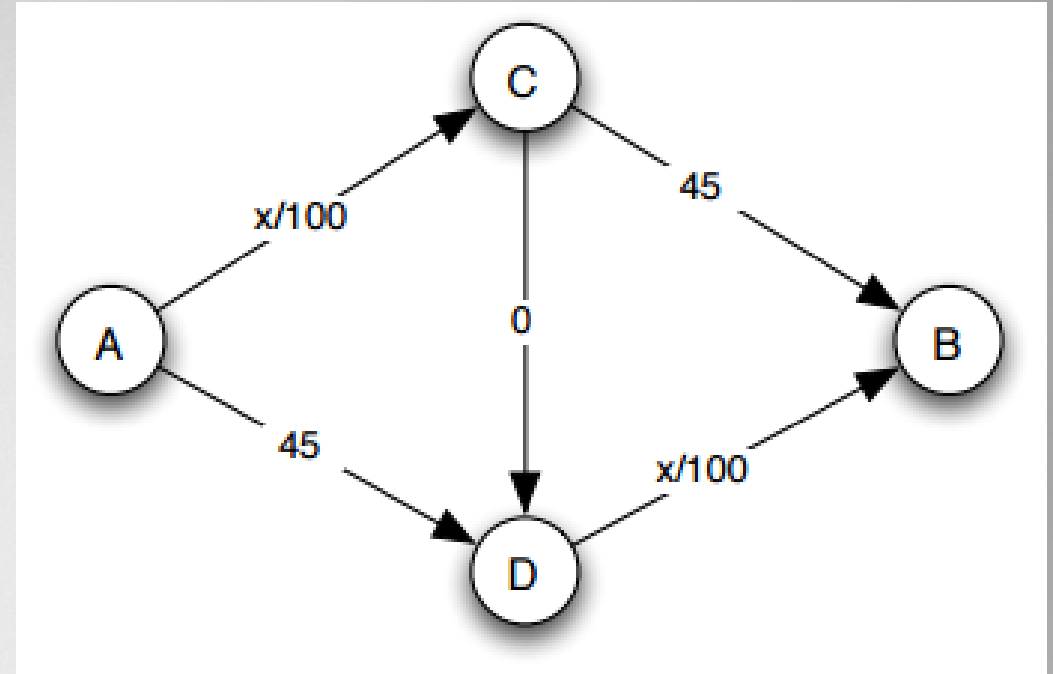
Traffic equilibrium

- ▶ Let h and w be the traffic flow (number of cars) through ACD and ADB respectively.
- ▶ Consider the following basic relationship: if a node receives certain amount of traffic flow, then the same amount must go out. This is called flow conservation.
- ▶ Thus that is why the traffic flow for edge CB is h and the traffic flow for edge DB is w
- ▶ We also know that $h+w=4000$, i.e. the total amount of cars.
- ▶ To characterize the equilibrium, we must equate total travel time among the routes given flows h and w .
- ▶ Travel time for ACB: $h/100+45$
- ▶ Travel time for ADB: $45+w/100$
- ▶ Equating them: $h/100+45=45+w/100$ which yields the equation $h=w$.
- ▶ Since we also know that $h+w=4000$, then $h=w=2000$.
- ▶ Then a Nash equilibrium traffic pattern is for 2000 cars to go through route ACB and 2000 cars to go through route ADB. Total travel time for each route is 65.
- ▶ There are many equilibria: basically, any specification of a travel route for each driver such that 2000 of them go through ACB and the other 2000 go through ADB is an equilibrium.



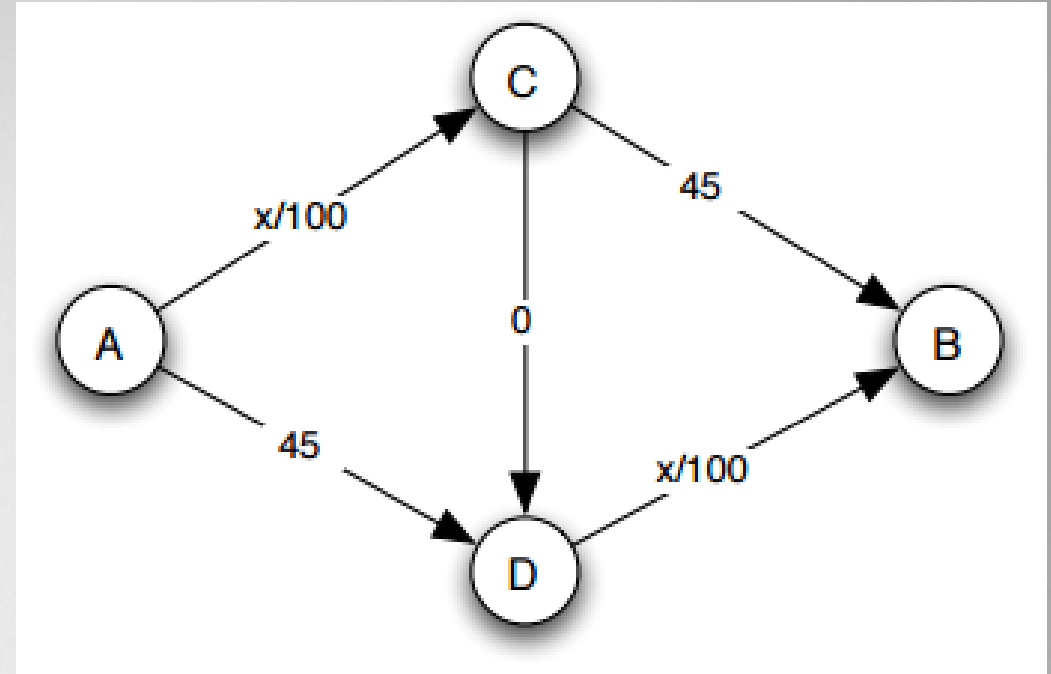
Braess' Paradox

- ▶ In the previous case, everything works out very cleanly, at an equilibrium, the self-interested behavior by all drivers causes them to balance perfectly between the available routes.
- ▶ However, by adding one more route to the network, there can be a drastic change...
- ▶ Suppose that the city government decides to build a fast highway between C and D.
- ▶ For simplicity, assume that it has a travel time of 0.
- ▶ Adding this new route would certainly help to improve driving times, right?
- ▶ Recall our discussions in the previous Module: sometimes *the more is not the better*.



Braess' Paradox

- ▶ Now the drivers have three routes to choose from: ACB, ADB, which have travel times $x/100+45$ both and the new route ACDB which has travel time $2x/1000=x/50$
- ▶ Notice that $x/50 < x/100+45$ if and only if $x < 4500$.
- ▶ Since there are only 4000 the previous holds.
- ▶ Thus, route ACDB is a **dominant strategy**, no matter how many cars are going through this route, a driver gains by switching the route through C and D.
- ▶ But if all 4000 cars take this route, what is the travel time?
- ▶ The travel time through route ACDB is $4000/50=80$.
- ▶ Notice that there is no optimal deviation, a driver considering taking either routes ACB or ADB would make a travel time of 85 minutes by switching.
- ▶ Once the fast highway is built through C and D, it acts like a vortex that draws all drivers into it, to the detriment of all.
- ▶ The phenomenon of “adding resources to a transportation network can **sometimes** hurt performance” was first described by Dietrich Braess in 1968.
- ▶ Given our knowledge at this point of Game Theory, we can see that this is an instance of a Prisoners' Dilemma, so is not actually that paradoxical isn't it?



Braess' Paradox: Real life

- ▶ This excerpt is taken from an article in the New York Times, published in December 25, 1990 ([link](#)):

ON Earth Day this year, New York City's Transportation Commissioner decided to close 42d Street, which as every New Yorker knows is always congested. "Many predicted it would be doomsday," said the Commissioner, Lucius J. Riccio. "You didn't need to be a rocket scientist or have a sophisticated computer queuing model to see that this could have been a major problem."

But to everyone's surprise, Earth Day generated no historic traffic jam. Traffic flow actually improved when 42d Street was closed.

- ▶ In the early two-thousands, when city planners in Seoul, South Korea, replaced a six lane highway with a five mile long park, traffic flow improved. In 1969, congestion in Stuttgart, Germany improved only after a newly built section of road was shut down ([link](#)).
- ▶ Braess's paradox doesn't tell us we should stop building more roads. Cars need roads to get from point A to point B. What it does tell us is that adding more roads doesn't always make things better. Sometimes less really is more.



NYC DOT

This "before" view of Times Square shows a streetscape that is not very friendly to pedestrians and bicyclists.



NYC DOT

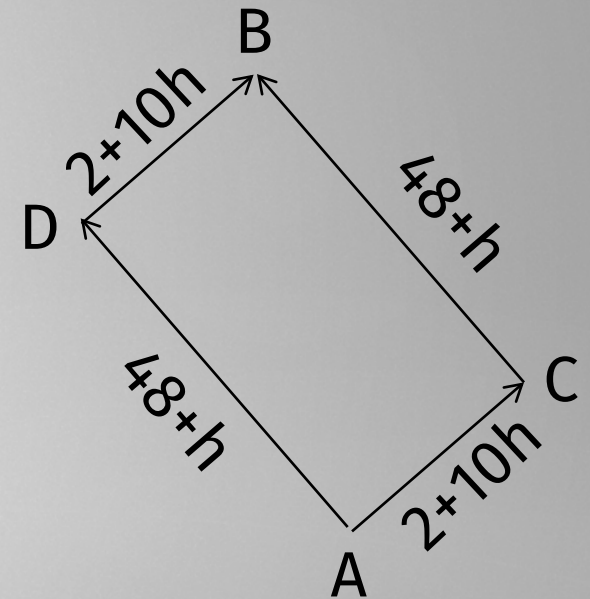
This "after" shot of Times Square shows an area that is inviting to residents and visitors alike.

Braess' Paradox

New York's Mayor Bloomberg spearheaded experimental road closures in 2009. At issue were two sections of Broadway where it obliquely crosses major north/south boulevards at Times and Herald Squares. The experiment was pitched as a way to reduce congestion. Projections were high, with travel times estimated to fall up to forty percent on some streets. When the numbers came in they weren't as lofty as advertised. But overall congestion appeared to have dropped. The experiment was deemed a success and the road closures were made permanent. Times and Herald Squares are now home to successful pedestrian plazas — plazas that haven't exacerbated traffic headaches

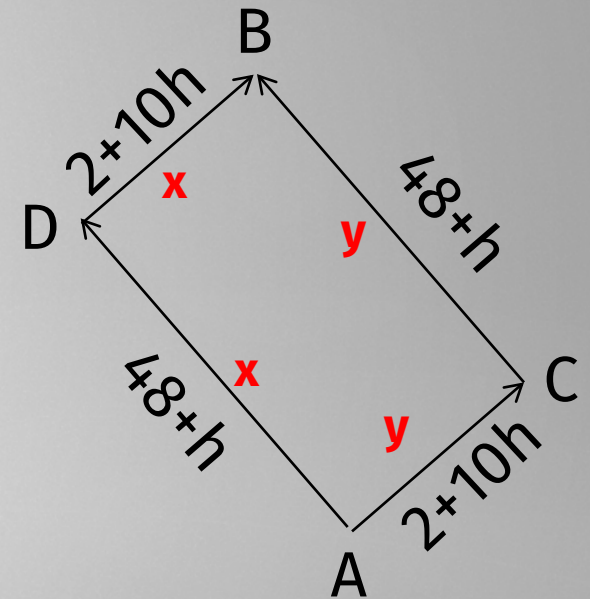
More complicated example

- ▶ Consider a road network as shown in the next diagram, on which 6,000 drivers travel from point A to point B
- ▶ The travel time in minutes on AC road is $2+10h$ where h is the number of thousands of cars on this road (if the number of cars on AC is 2,000 then the travel time from A to C is 22 minutes)
- ▶ Likewise the travel time on BD is also $2+10h$
- ▶ The travel time on CB or AD is $48+h$
- ▶ What is the travel time of a car that goes from A to B when the system is in equilibrium (meaning that traveling through C takes the same time as through D)



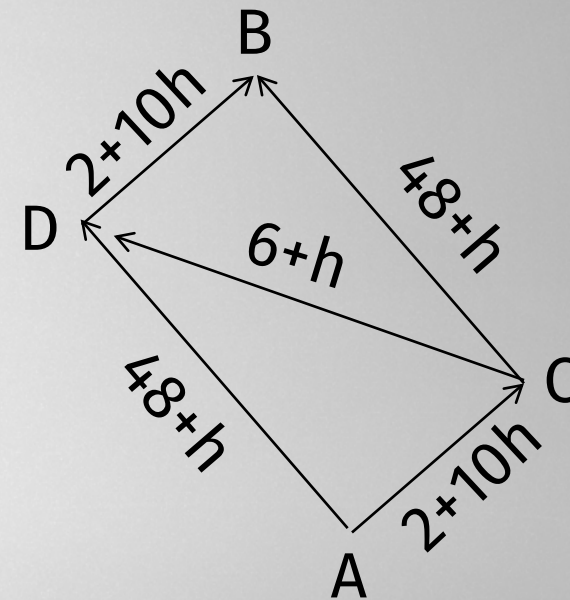
More complicated example

- ▶ Again, say that the flow that departs from node A to C is y and the one that departs from A to D is x . Flows are in thousands.
- ▶ Flow conservation then requires that the flow through road DB is also x and the flow through CB is y .
- ▶ Furthermore, since 6000 cars depart from A, the same amount of cars must reach B. Thus $x+y=6$.
- ▶ For the traffic pattern to be at equilibrium, travel times must be equal:
 - ▶ Travel time ADB: $50+11x$
 - ▶ Travel time ACB: $50+11y$
 - ▶ Equating them yields $x=y$
 - ▶ $x=y$ together with $x+y=6$ means that $x=y=3$.
 - ▶ Total travel time through either route is 83 minutes.

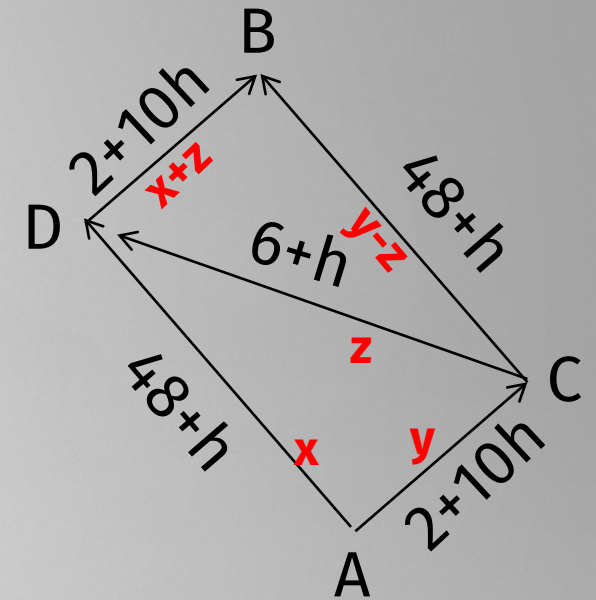


More complicated example

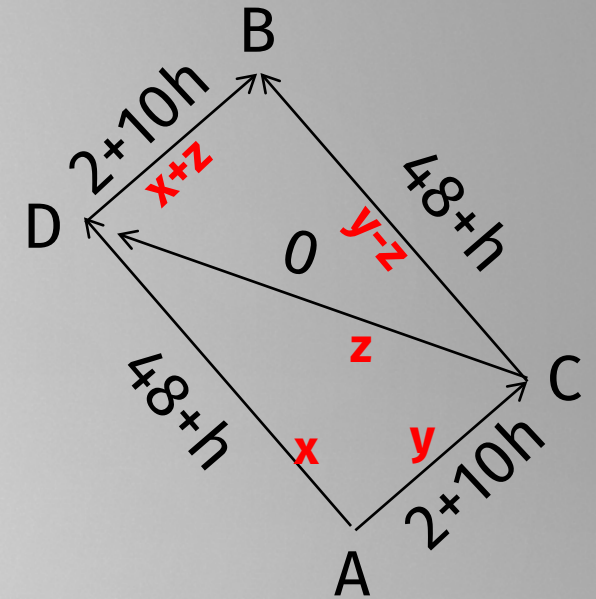
- ▶ Now suppose that the two points C and D are connected through a relatively short road and the travel time on CD is $6+h$ (for simplicity assume that the road CD is one way and one can only travel from C to D)
- ▶ What is now the equilibrium travel time?
- ▶ Notice that there are no dominant strategies now.



- ▶ Let's name the flow that departs from A to D as x and the one that departs from A to C as y . At C, flow y bifurcates, the amount z goes through CD and the amount $y-z$ goes through CB. Note D is reached by flow x and flow z , thus from D, flow $x+z$ departs towards B. Furthermore, we know that $x+y=6$. In setting up this, remember we used *flow conservation*.
- ▶ Now we can calculate travel times for the 3 routes:
 - ▶ ADB: $50+11x+10z$
 - ▶ ACB: $50+11y-z$
 - ▶ ACDB: $10+10x+10y+11z$
 - ▶ We will equate the travel time ADB with ACB, and the travel time of ACB with ACDB to get two equations:
 - ▶ $11x-11y+11z=0$
 - ▶ $10x-y+12z=40$
 - ▶ Together with $x+y=6$, these forms a system of three equations in 3 unknowns.
 - ▶ The solution is: $x=2, y=4, z=2$.
 - ▶ Total travel time through each route is 92, which is worse than before.

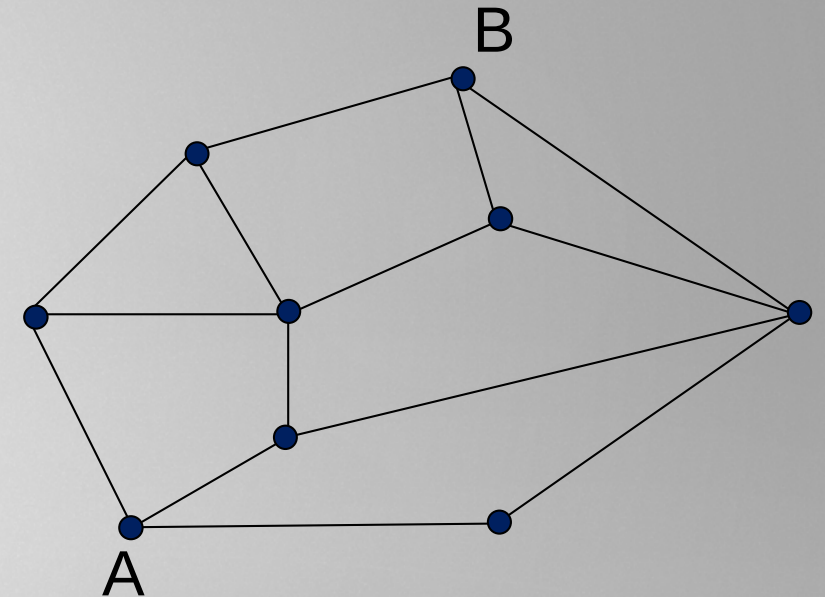


- ▶ Suppose now that you can drive from C to D in no time, what will happen now?
- ▶ We calculate travel times for the 3 routes:
 - ▶ ADB: $50+11x+10z$
 - ▶ ACB: $50+11y-z$
 - ▶ ACDB: $4+10x+10y+10z$
 - ▶ We will equate the travel time ADB with ACB, and the travel time of ACB with ACDB to get two equations:
 - ▶ $11x-11y+11z=0$
 - ▶ $10x-y+11z=40$
 - ▶ Together with $x+y=6$, these forms a system of three equations in 3 unknowns.
 - ▶ The solution is: $x=14/11$, $y=52/11$, $z=38/11$.
 - ▶ Total travel time through each route is 98.5454, which is even worse than before.



Communication networks

- ▶ Communication networks can be modelled in the same way as traffic networks.
- ▶ In a communication network, say e-mail, every bit travel from A to B through some nodes of the net.
- ▶ In each node there is a processor that takes an independent decision as to where to send each bit (like the driver who makes his own decision as to what road to take)
- ▶ In this case more links may sometimes hurt the system
- ▶ A careful design of the network is critical.
- ▶ There is also the question of how does the system works under the assumption of independent decision-makers versus a centralized system.
- ▶ A centralized system solves the flow problem by calculating the minimal total cost flow for the traffic network. This would be the *social optimum*.
- ▶ *Comparing the social optimum versus the equilibrium (decentralized) total cost gives an idea of what is the price of letting the system function independently: "the price of anarchy".*
- ▶ This is an active area of research in computer science, if you are interested.



Location Games

Location games (or where to put my ice-cream shop)

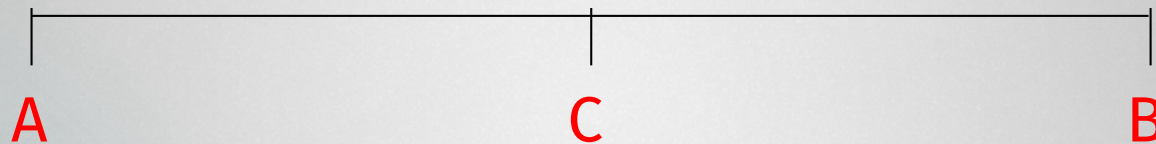
- ▶ Two service providers have to choose their locations.
- ▶ The possible locations are corners on a certain straight street.
- ▶ Customers are **uniformly distributed** along the street and each one of them approaches the **closest** provider.
- ▶ If the two providers choose the same location each serves half of the customers



www.alamy.com - EYATFD

Location games: Example 1,two providers

- ▶ Suppose that there are three possible locations A, B and C where A and B are the two ends of the street and C is in the middle. Where would the two providers be located?



- ▶ Solution:
 - ▶ It is a strictly dominated strategy for each provider to choose either A or B. The location C strictly dominates both locations A and B. Therefore, both providers will choose the location C and each one of them will serve half of the customers.
 - ▶ To see this, suppose that both providers are located respectively at A and B.
 - ▶ Each one will serve half the market. But consider the provider located at A, he reasons: if I move to C, then I will serve costumers located in the interval AC and also steal some consumers in the interval CB that B was previously serving.
 - ▶ Thus, A has an optimal deviation to move to C. Similarly for B.
 - ▶ They both end up serving again half of the market.

Location games: Example 2, two providers

- ▶ As in Problem 1 but now there are five possible locations with equal distance between any two adjacent locations.



- ▶ Solution:
 - ▶ The locations A and B are both strictly dominated by the location D. Thus, these two locations can be eliminated.
 - ▶ After eliminating A and B, the locations C and E are strictly dominated by D. Thus, the two providers will choose location D and they will again split the market equally.



- ▶ With the same argument (successive elimination of strictly dominated strategies) we conclude that whenever there are odd number of possible locations with equal distances, the two providers choose the middle location and each serves half of the market.

Location games: Example 3, two providers

- ▶ As in the first example, but now there are two possible locations:



- ▶ Solution:
 - ▶ Since consumers are located uniformly in the interval AB, when each provider takes one of the positions, each serves half of the market.
 - ▶ No matter where the providers are located, each will serve half of the consumers.

Location games: Problem 4, two providers

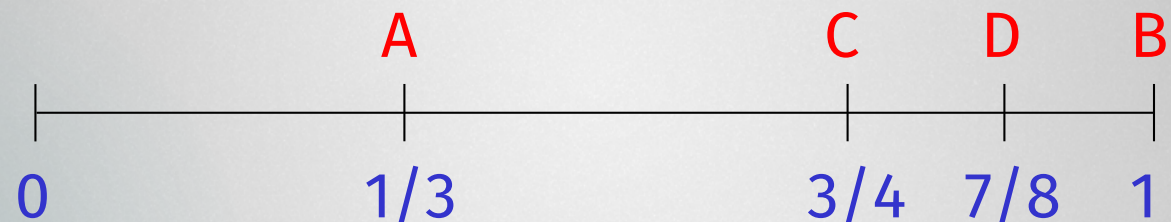
- ▶ As in Problem 1 but now there are four possible locations:



- ▶ Solution:
 - ▶ The location A is strictly dominated by the location C and the location B is strictly dominated by the location D.
 - ▶ To see this, notice that if whenever the other provider is located, choosing A is always dominated by C or D, because in that way, the provider gets a larger share of the market. Similarly for location B.
 - ▶ After eliminating both locations A and B any of the other two locations C or D are equally good for both providers and each one of them will serve half of the market.
 - ▶ Using successive elimination of strictly dominated strategies **we conclude that whenever the number of locations is even then with equal distances the middle two locations** are the only equilibrium points. In the four location case the four Nash equilibrium points are (C,C), (C,D), (D,C) and (D,D).

Two providers with finite number of locations and arbitrary distances

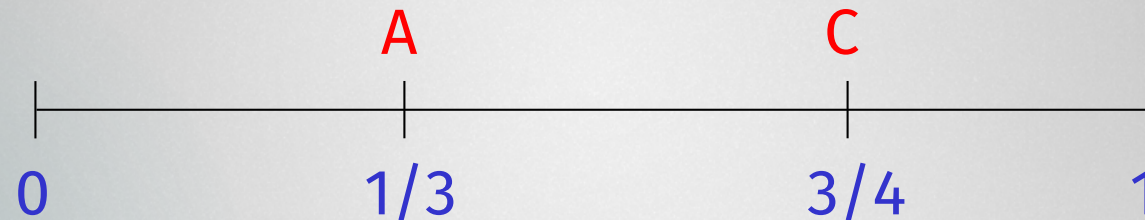
- ▶ Consider the case where customers are uniformly distributed on the interval $[0, 1]$. Suppose that there are four possible locations: A, B, C and D as in the following figure:



- ▶ Where would the two providers be located?
- ▶ Solution:
 - ▶ First note that the location B is strictly dominated by the location A.
 - ▶ After eliminating B, the location D becomes strictly dominated by A.

Two providers with finite number of locations and arbitrary distances

- After eliminating D we have:



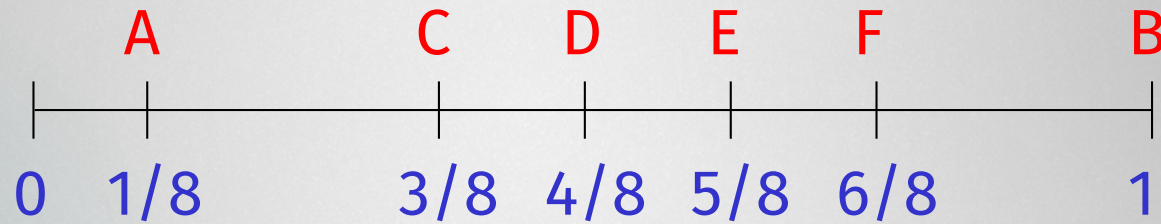
- Suppose that Provider 1 is located at C. If Provider 2 is also located at C, then Provider 1 obtains $\frac{1}{2}$ and if he moves to A then he obtains $\frac{1}{3} + \frac{1}{2}(\frac{3}{4} - \frac{1}{3}) = \frac{13}{24} > \frac{1}{2}$. If Provider 2 is located at A, then Provider 1 who is located at C obtains $\frac{1}{4} + \frac{1}{2} * (\frac{3}{4} - \frac{1}{3}) = \frac{11}{24}$. Hence he is better off moving to A to obtain $\frac{1}{2}$. **Thus the two providers should choose A.**

Two providers-The general case.

- ▶ Suppose that there are two providers with n possible locations and with arbitrary distances between locations. Then
- ▶ Claim:
 - ▶ the only location that survives successive elimination of strictly dominated strategies is the one that is the closest to the **median customer** (The customer located at the middle of the street). If there are two such locations (as in Problem 3 and 4) both of them are equally good for the providers. **The two providers split the market equally** (whether they are located in the same corner or in two adjacent corners).
- ▶ The proof of the claim is along the same lines as the solution of the previous examples, but I won't present it here.
- ▶ Let's look at another example.

Two Providers- The General Case

- ▶ Example: Consider the following situation:



- ▶ D is the location of the median customer and hence the two providers will choose D as their location.

Three or more providers

- ▶ Consider three providers with two locations (the end points):



- ▶ With three or more provider we do not have dominated strategies and the previous procedures of successive elimination of dominated strategies can not be applied. The solution concept we employ here is hence the Nash equilibrium.
- ▶ Is it a Nash equilibrium for all three to be located at the same point, say A?
- ▶ The answer is no. Suppose all three providers are located at A. Each obtains $1/3$ of the market. Any one of them who unilaterally deviates to B will obtain $1/2$ of the market and therefore has incentive to do so.

Three or more Providers

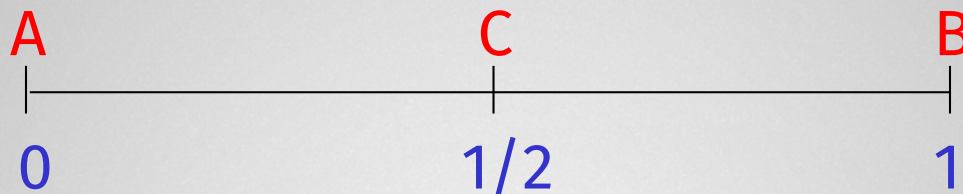
- Suppose that two providers choose A and the other one chooses B:



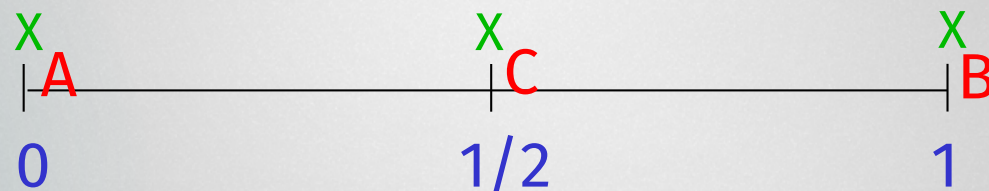
- Certainly, the one in B who obtains $1/2$ has no incentive to deviate to A since in A he will obtain $1/3$ only. Also, no one at A can benefit from switching to B (in both cases the deviant will obtain $1/4$). We conclude that the only equilibrium outcome is: two providers are located in A and the other provider is located in B or vice a versa.

Three or more providers

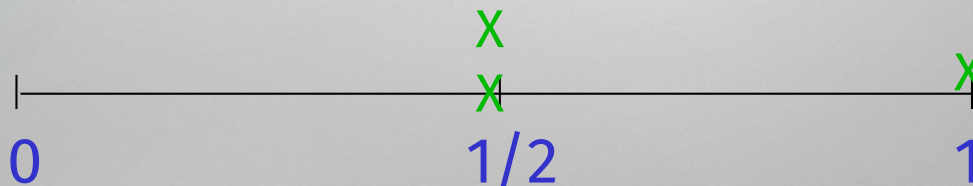
- Next suppose there are three locations A, B and C with three providers, as in the next figure:



- Is the following an equilibrium outcome?

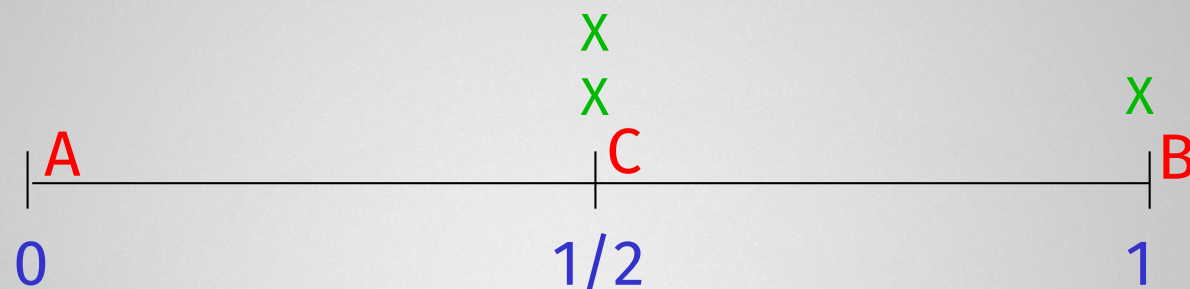


- Answer:
 - No! The provider in A who serves $1/4$ of the market is better off deviating to C. He will then obtain $1/2 * (1/2 + 1/4) = 3/8 > 1/4$



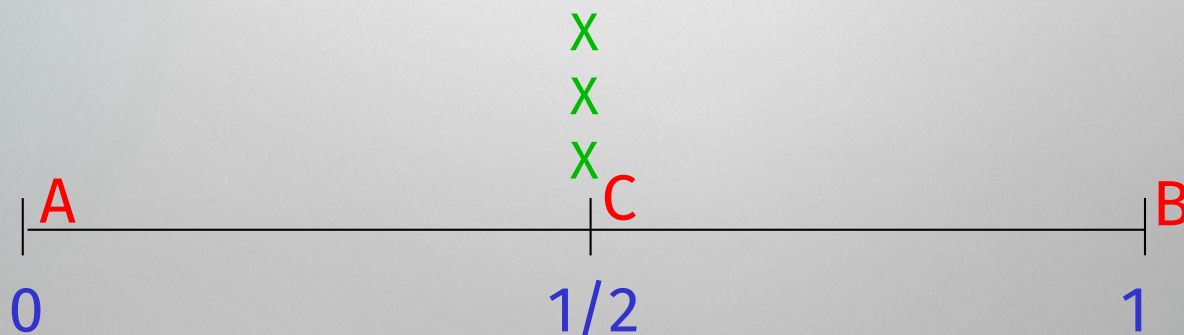
Three or more providers

- Is the following an equilibrium outcome?



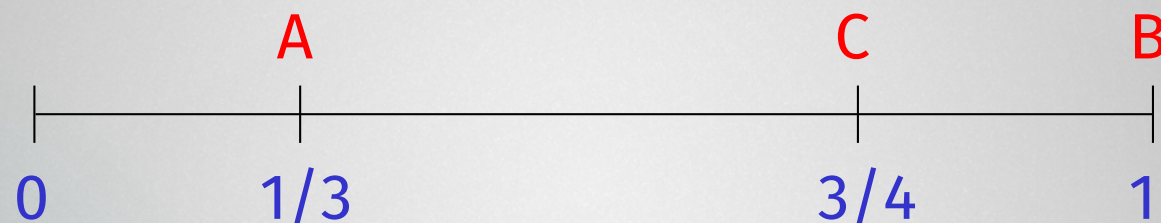
- Answer:

- No! the provider in B who serves $1/4$ of the market is better off deviating to C where he will serve $1/3$ of the market.
- The only equilibrium is that the three providers choose their location at C:



Exercise (for you to actually try)

- ▶ Consider the following three providers-three locations problem:



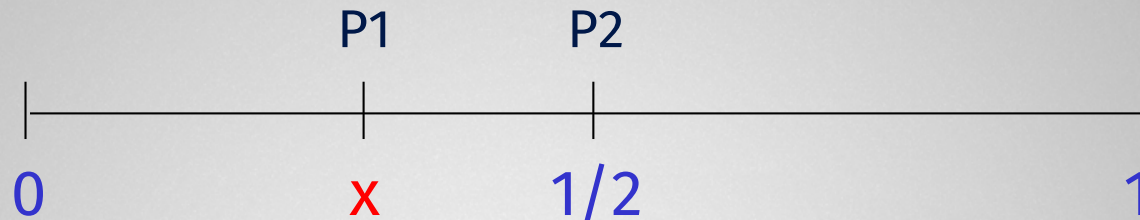
- ▶ Check that the only equilibrium outcome is that two providers choose A and the other provider chooses C

No restriction on locations

- ▶ Suppose next that a provider can choose any point in $[0, 1]$ as his location (continuum of possible locations). Let us start with two providers.
- ▶ Claim:
 - ▶ The only equilibrium with two providers is when both provider choose the middle point $1/2$, as their location.
- ▶ Proof:
 - ▶ First let us show that choosing the middle location for both of the providers constitutes a Nash equilibrium. If they both choose $1/2$ as their location each serves half of the market.

No restriction on locations

- Suppose that one provider deviates to x , $x < 1/2$:

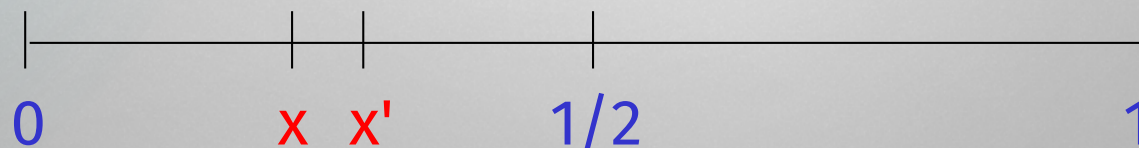


- The provider at x obtains $[(x + 1/2)/2]$. Since $x < 1/2$ he obtains less than $1/2$ and he loses from this deviation. Similar outcome is obtained if he deviates to $x > 1/2$.

- We conclude that choosing the middle point $1/2$ as the location of both providers is a Nash equilibrium.

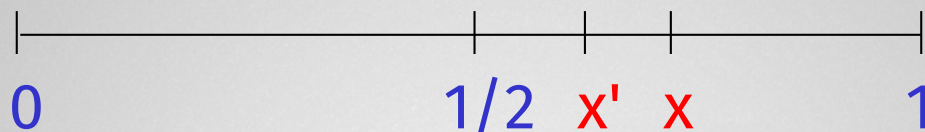
- Let us next show that there is no other Nash equilibrium. Suppose first that the two providers choose the same location x and $x \neq 1/2$. Each then obtains $1/2$.

- If $x < 1/2$ then a provider who slightly deviates to the right of x , say to x' , obtains more than $1/2$ (actually $1 - x' + (x' - x)/2$) and will benefit from this deviation.



No restriction on locations

- If $x > 1/2$ a beneficial deviation is to the left of x . A deviant to x' will obtain $(x' + (x - x')/2) > 1/2$:

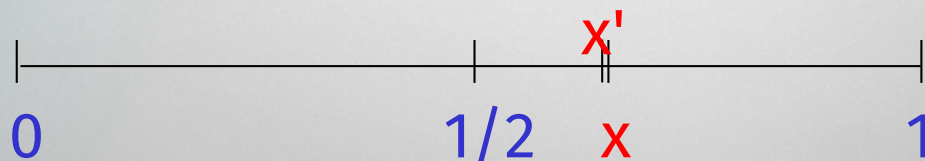


- We conclude that choosing the same location but other than $1/2$ is not a Nash equilibrium. We are left with the case where the two providers choose different locations, say x and x' , $x \neq x'$.

- Suppose that $x < 1/2$:



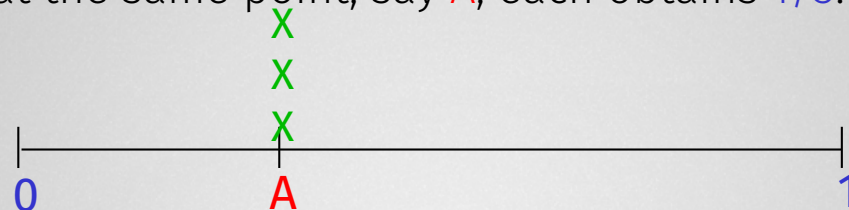
- then the provider on x' benefits from moving to a location very close to x and to the right of x . if $x > 1/2$:



- then the provider on x' has incentive to move to a point left of x and very close to x .

No restriction on locations

- ▶ What happens if there are **three** providers?
- ▶ There is no (pure strategy) Nash equilibrium in this case! why?
- ▶ If all three are located at the same point, say **A**, each obtains $1/3$.



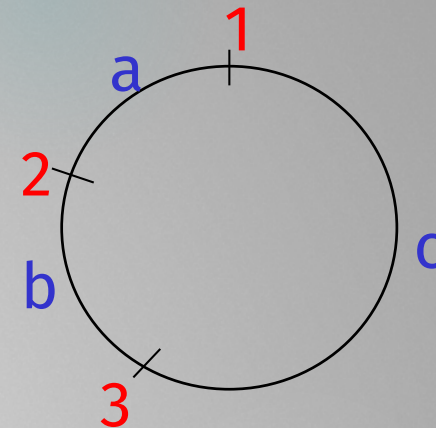
- ▶ Suppose that $A \leq 1/2$. Then a deviant to a point on the right of **A** but close to **A** will increase his payoff from $1/3$ to a payoff very close to $1 - A \geq 1/2$.
- ▶ Suppose next that not all providers are located at the same point. Then there is at least one provider who is located alone at say **A** and there is no other provider between **A** and either **0** or **1**:



- ▶ By slightly moving to the right the provider in **A** will increase his payoff and hence has an incentive to deviate.
- ▶ There is a mixed strategy equilibrium, but it involves mixing over a continuum of strategies (the location in $[0,1]$). We will not go over it since it is a little advanced.

A circular street

- ▶ The customers are uniformly distributed on a circle of length (circumference) 1. There are three providers.



- ▶ The customers are uniformly distributed on a circle of length (circumference) 1. There are three providers.

- ▶ Solution:

- ▶ We claim that the set of all Nash equilibrium points consists of three locations 1, 2 and 3
- ▶ with distances a , b and c (see Figure) such that $a \leq 1/2$, $b \leq 1/2$, $c \leq 1/2$ and $a+b+c=1$.
- ▶ Note that provider 1 obtains $(a+c)/2$ and this does not change if 1 moves toward either 2 or 3. Hence, he may only benefit if he deviates to a point in between 2 and 3. In this case he obtains $b/2$. Therefore in equilibrium we must have: $(a+c)/2 \geq (b/2)$ or equivalently $a+c \geq b$. Since $a+c+b=1$ we have $1-b \geq b$ or $b \leq 1/2$. By similar argument (taking into account the deviation of 2 or 3) we have $a \leq 1/2$ and $c \leq 1/2$

Political contests

ELECTORAL COMPETITION AND THE MEDIAN VOTER

Political contests

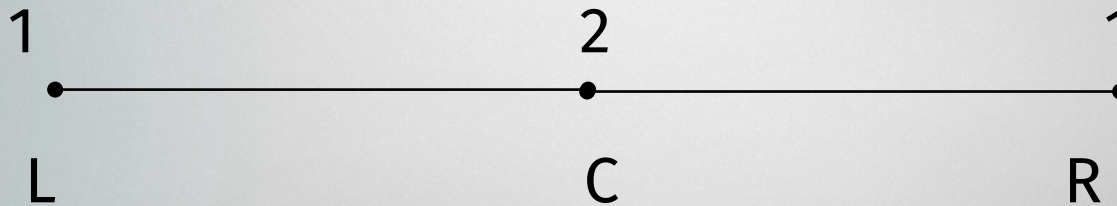
- ▶ Two political candidates compete for a certain job (president, prime-minister, etc.,)
- ▶ They are completely opportunistic and their only target is to win the job.
- ▶ Suppose first that there are three relevant positions in the political spectrum:

left (L), center (C), and right (R)

- ▶ Candidates choose position strategically and voters vote for the closest candidate to their position
- ▶ If both candidates are equally close to a voter, then he chooses one at random to vote for him.
- ▶ A candidate with the highest number of votes wins the job.
- ▶ If they obtain the same number of votes, then one of them is selected at random.

Example 1:

- ▶ Suppose that there are 4 million voters:
 - 1million of them prefer R
 - 2 million of them prefer C
 - other 1 million voters prefer L.
- ▶ What position each candidate choose?
- ▶ We will construct two tables to help analyze this situation.



Example 1: tables

Allocation of voters

- In the first, we have the allocation of the 4 million voters to candidates as a result of the candidates' strategic choices.

1 \ 2	L	C	R
L	2,2	1,3	2,2
C	3,1	2,2	3,1
R	2,2	1,3	2,2

Outcome

- The second one describes the outcome again as the result of the strategy choice of the two candidates
- 1 for winner and 0 for the loser, and $(\frac{1}{2}, \frac{1}{2})$ if both have the same chance of winning

1 \ 2	L	C	R
L	$\frac{1}{2}, \frac{1}{2}$	0,1	$\frac{1}{2}, \frac{1}{2}$
C	1,0	$\frac{1}{2}, \frac{1}{2}$	1,0
R	$\frac{1}{2}, \frac{1}{2}$	0,1	$\frac{1}{2}, \frac{1}{2}$

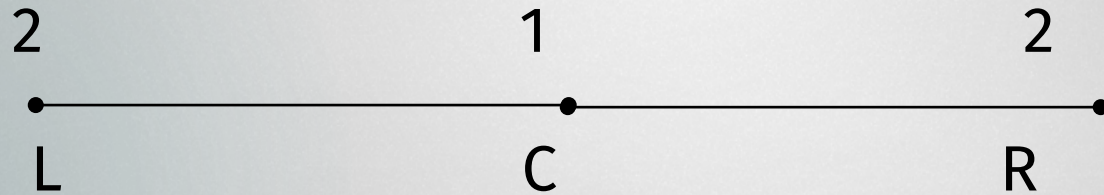
Example 1: solution

- Note that C is a strictly dominant strategy for both candidates. Consequently, they both will position themselves in the center and they will have equal chance to win.

		2		
1 \ 2		L	C	R
	L	$\frac{1}{2}, \frac{1}{2}$	0,1	$\frac{1}{2}, \frac{1}{2}$
	C	1,0	$\frac{1}{2}, \frac{1}{2}$	1,0
	R	$\frac{1}{2}, \frac{1}{2}$	0,1	$\frac{1}{2}, \frac{1}{2}$

Example 2

- ▶ Suppose that there are 2 candidates and 5 million voters:
 - 2 million of them prefer L
 - 1 million of them prefer C
 - 2 million voters prefer R.



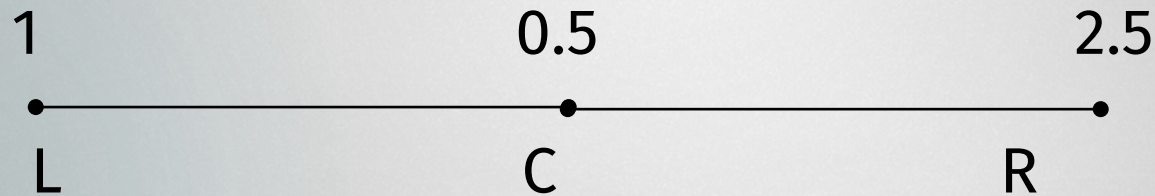
Example 2: solution

- ▶ Again C is a strictly dominant strategy and each will position himself in the center.

		2		
1 \		L	C	R
	L	$\frac{1}{2}, \frac{1}{2}$	0,1	$\frac{1}{2}, \frac{1}{2}$
	C	1,0	$\frac{1}{2}, \frac{1}{2}$	1,0
	R	$\frac{1}{2}, \frac{1}{2}$	0,1	$\frac{1}{2}, \frac{1}{2}$

Example 3

- ▶ Suppose that there are 2 candidates and 4 million voters:
 - 1 million of them prefer L
 - 0.5 million of them prefer C
 - 2.5 million voters prefer R.



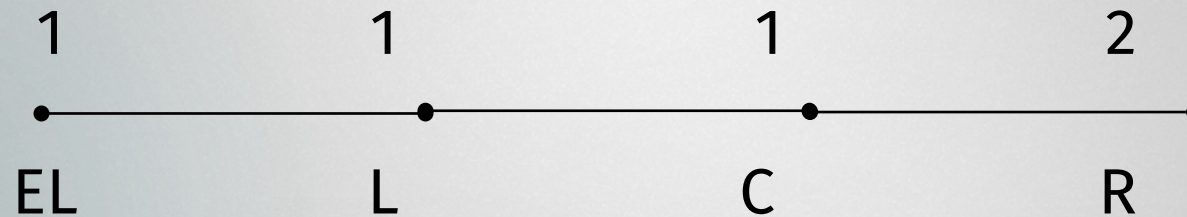
Example 3: solution

- R is a strictly dominant strategy for each candidate. The two candidates will position themselves on the Right

		2		
		L	C	R
1	L	$\frac{1}{2}, \frac{1}{2}$	0,1	0,1
	C	1,0	$\frac{1}{2}, \frac{1}{2}$	0,1
	R	1,0	1,0	$\frac{1}{2}, \frac{1}{2}$

Example 4

- ▶ Suppose there are four positions:
EL (extreme left), L, C and R
- ▶ And there are 5 million voters as follows



Example 4

- The corresponding distribution of voters is

1 \ 2	EL	L	C	R
EL	$2\frac{1}{2}, 2\frac{1}{2}$	1, 4	$1\frac{1}{2}, 3\frac{1}{2}$	2, 3
L	4, 1	$2\frac{1}{2}, 2\frac{1}{2}$	2, 3	$2\frac{1}{2}, 2\frac{1}{2}$
C	$3\frac{1}{2}, 1\frac{1}{2}$	3, 2	$2\frac{1}{2}, 2\frac{1}{2}$	3, 2
R	3, 2	$2\frac{1}{2}, 2\frac{1}{2}$	2, 3	$2\frac{1}{2}, 2\frac{1}{2}$

Example 4

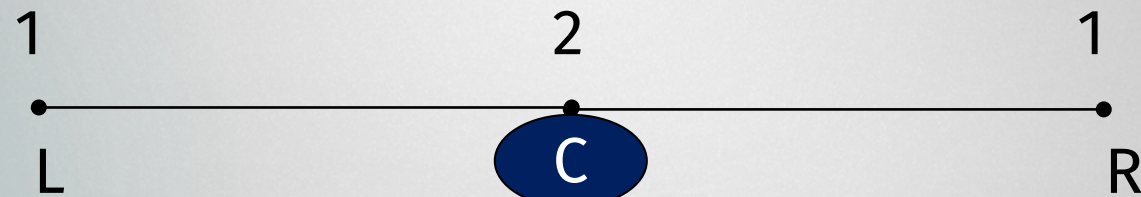
- ▶ The strategic game is

1 \ 2		EL	L	C	R
1	EL	1/2, 1/2	0, 1	0, 1	0, 1
	L	1, 0	1/2, 1/2	0, 1	1/2, 1/2
	C	1, 0	1, 0	1/2, 1/2	1, 0
	R	1, 0	1/2, 1/2	0, 1	1/2, 1/2

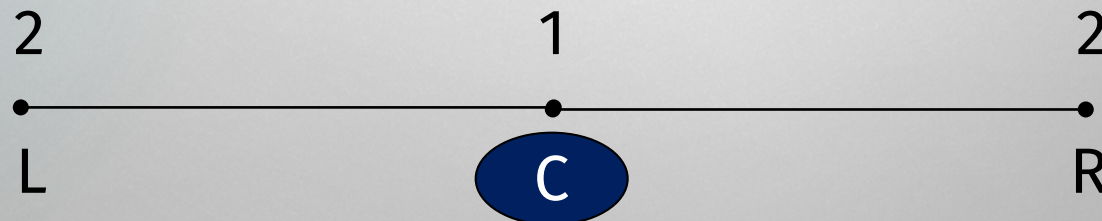
- ▶ Here C is a (weakly) dominant strategy for each candidate. Note also that C can be obtained by successive elimination of strictly dominated strategies. Both candidate will be located in the Center.

What do we learn from the previous examples?

- ▶ A median position is a position where at least half of the voters prefer that position or positions on the left of it and at least half of the voters prefers that position or positions on the right of it.
- ▶ Example 1: C is the median position.

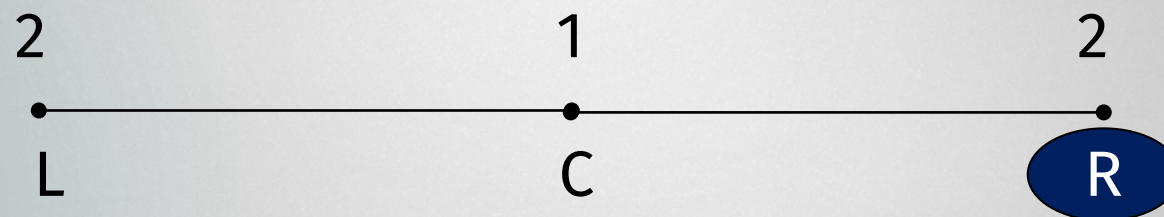


- ▶ Example 2: C is again the median position.

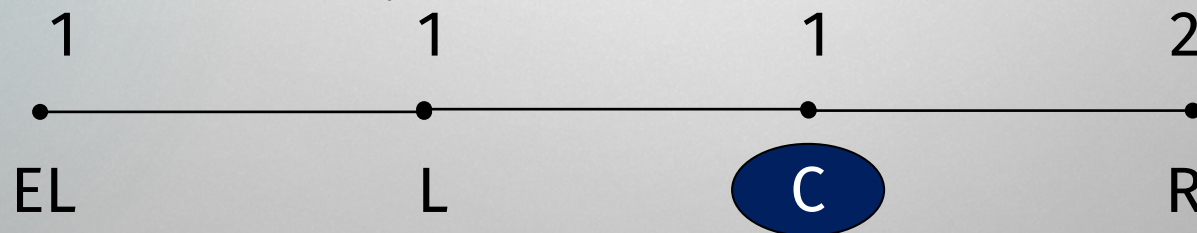


What do we learn from the previous examples?

- ▶ A median position is a position where at least half of the voters prefer that position or positions on the left of it and at least half of the voters prefers that position or positions on the right of it.
- ▶ Example 3: R is the median position.



- ▶ Example 4: C is the median position.



The median position

- ▶ Observation:
 - ▶ In all these examples the median position is a strictly dominant strategy for both candidates, or can be obtained by successive elimination of strictly dominated strategy.
- ▶ Proposition:
 - ▶ *Consider two candidates with any number of position that can be ranked from left to right. Suppose that the median position is unique. Then the median position is a weakly dominant strategy for any candidate and it can be obtained by successive elimination of strictly dominated strategies.*

Multiple Medians

- Consider 4 positions A, B, C and D, and 6 million voters, as follows:



- Here B and C are median positions.

Multiple medians

- The distribution of voters is

1 \ 2	A	B	C	D
A	3, 3	1, 5	2, 4	3, 3
B	5, 1	3, 3	3, 3	4, 2
C	4, 2	3, 3	3, 3	5, 1
D	3, 3	2, 4	1, 5	3, 3

Multiple medians

- The strategic game is

1 \ 2		A	B	C	D
1	A	1/2, 1/2	0, 1	0, 1	1/2, 1/2
	B	1, 0	1/2, 1/2	1/2, 1/2	1, 0
	C	1, 0	1/2, 1/2	1/2, 1/2	1, 0
	D	1/2, 1/2	0, 1	0, 1	1/2, 1/2

- Note that A and D are strictly dominated strategies and every candidate is indifferent between B and C. Therefore, each candidate will choose either the median position B or the median position C.

Multiple medians

- ▶ The strategic game is

1 \ 2		A	B	C	D
A		1/2, 1/2	0, 1	0, 1	1/2, 1/2
B		1, 0	1/2, 1/2	1/2, 1/2	1, 0
C		1, 0	1/2, 1/2	1/2, 1/2	1, 0
D		1/2, 1/2	0, 1	0, 1	1/2, 1/2

- ▶ There are four possible sensible outcomes:
(B,B), (B,C), (C,B) and (C,C)
- ▶ In each one of them, the two candidates have equal chance to win.